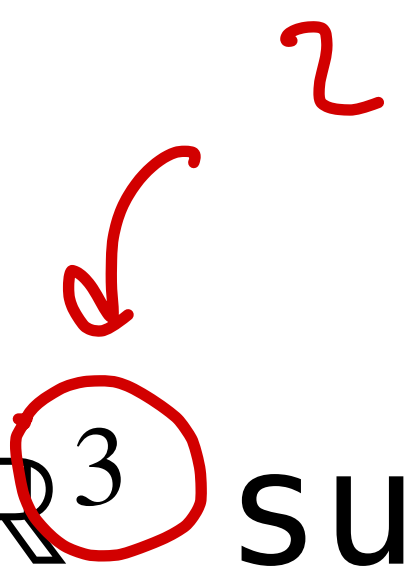


Linear Transformations

Geometric Algorithms
Lecture 7

Practice Problem

Find three vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ in $\mathbb{R}^3$ such that



» every pair of vectors $\{\mathbf{v}_1, \mathbf{v}_2\}$, $\{\mathbf{v}_1, \mathbf{v}_3\}$, $\{\mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent

» $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent

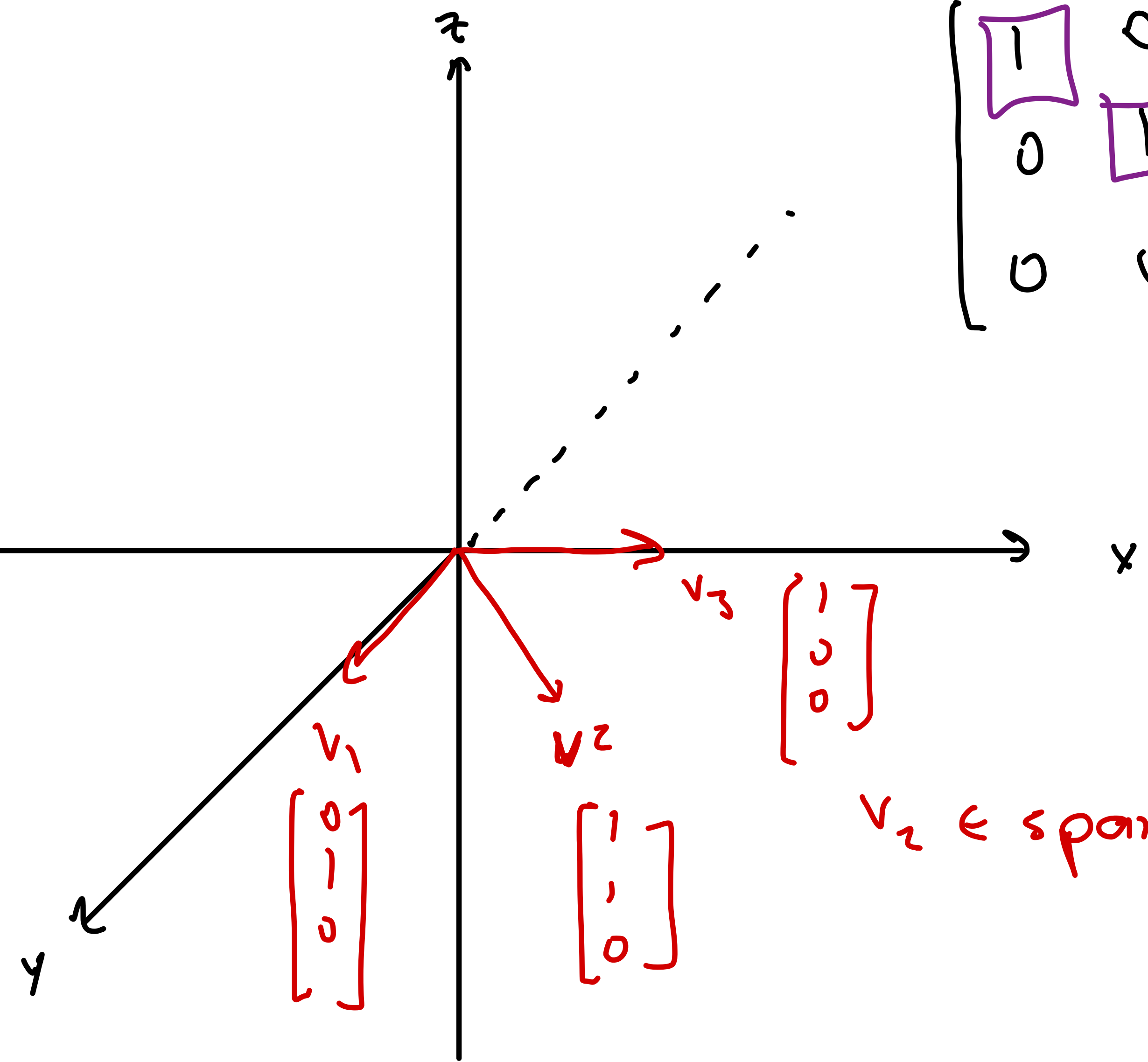
$$\begin{bmatrix} v_1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} v_2 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} v_3 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} v_1 & v_2 \\ \boxed{1} & 0 \\ 0 & \boxed{1} \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} v_2 & v_3 \\ 0 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix} \sim \begin{bmatrix} \boxed{1} & 1 \\ 0 & \boxed{1} \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} v_1 & v_3 \\ \boxed{1} & 1 \\ 0 & \boxed{1} \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} \boxed{1} & 0 & \cancel{1} \\ 0 & \boxed{1} & 1 \\ 0 & 0 & 0 \end{bmatrix}$$



$v_2 \in \text{span}\{\vec{v}_1, \vec{v}_3\}$

Objectives

- » Introduce matrix transformations
- » Define linear transformations
- » Start looking at the geometry of linear transformations

Keywords

Transformations

Domain, Codomain

Image, Range

Matrix Transformations

Linear Transformations

Additivity, Homogeneity

Dilation, Contraction, Shearing, Rotation

Matrix Transformations

Recall: Spans (with Matrices)

$$\text{span} \{ \vec{v}_1, \dots, \vec{v}_n \} = \{ \alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n : \alpha_1, \dots, \alpha_n \in \mathbb{R} \}$$

$$A \vec{u} = A \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} = [\vec{a}_1 \dots \vec{a}_n] \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} = u_1 \vec{a}_1 + \dots + u_n \vec{a}_n$$

$$\text{span} \{ \vec{v}_1, \dots, \vec{v}_n \} = \{ [\vec{v}_1 \dots \vec{v}_n] \vec{x} : \vec{x} \in \mathbb{R}^n \}$$

Recall: Spans (with Matrices)

Df. The *span* of a set of vectors is the set of all possible linear combinations

$$\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\} = \{[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] \mathbf{v} : \mathbf{v} \in \mathbb{R}^n\}$$

Matrices as Transformations

$$\vec{x} \mapsto Ax$$

$$T(\vec{x}) = \underbrace{A}_{\text{matrix}} \vec{x}$$

$m \times n$

$$m \begin{bmatrix} n \end{bmatrix}$$

Matrices as Transformations

Matrices *transform* vectors into vectors the span of its columns

$$\mathbf{x} \mapsto A\mathbf{x}$$

map a vector $\mathbf{v}$ to the vector $A\mathbf{v}$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

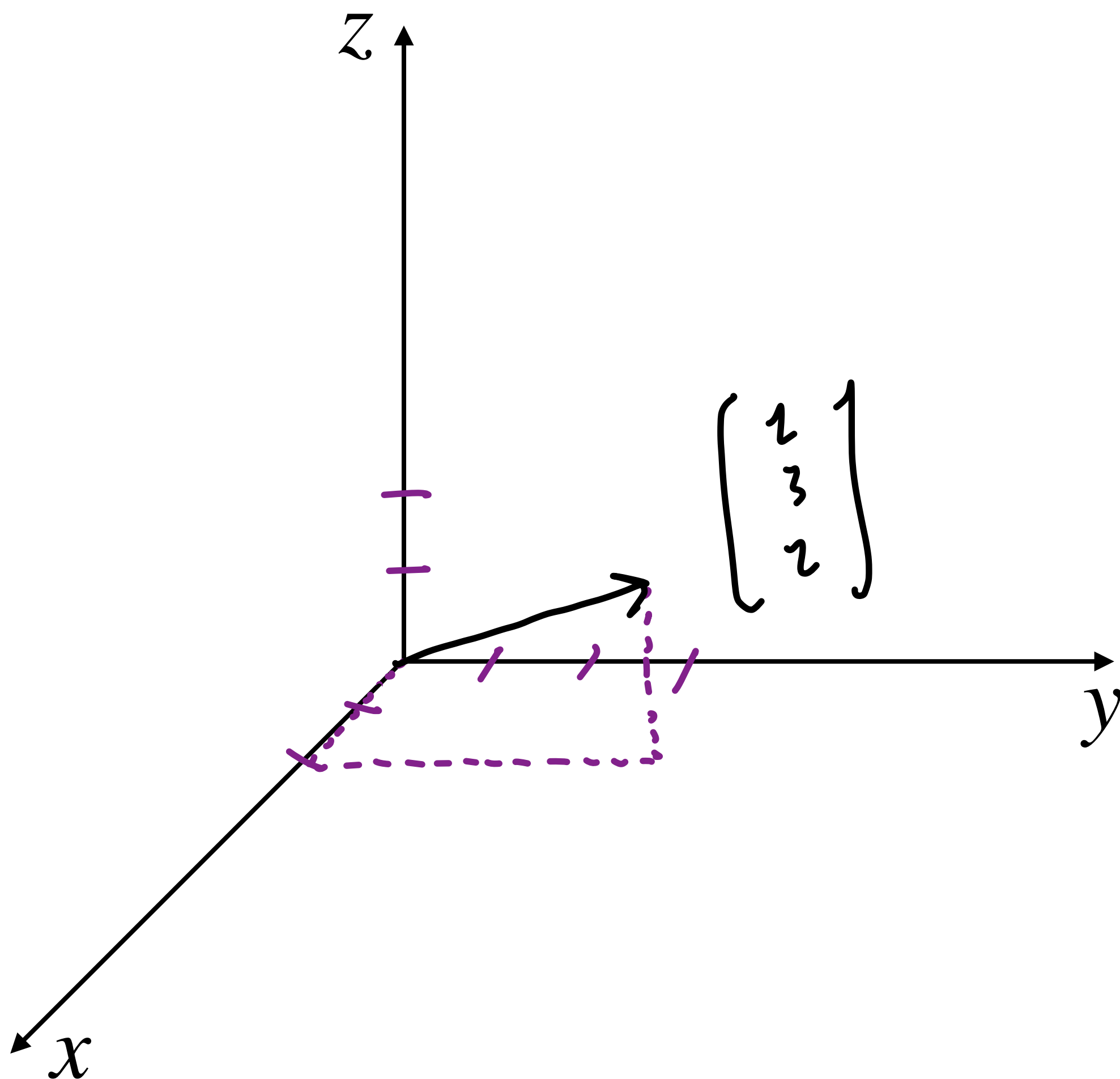
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

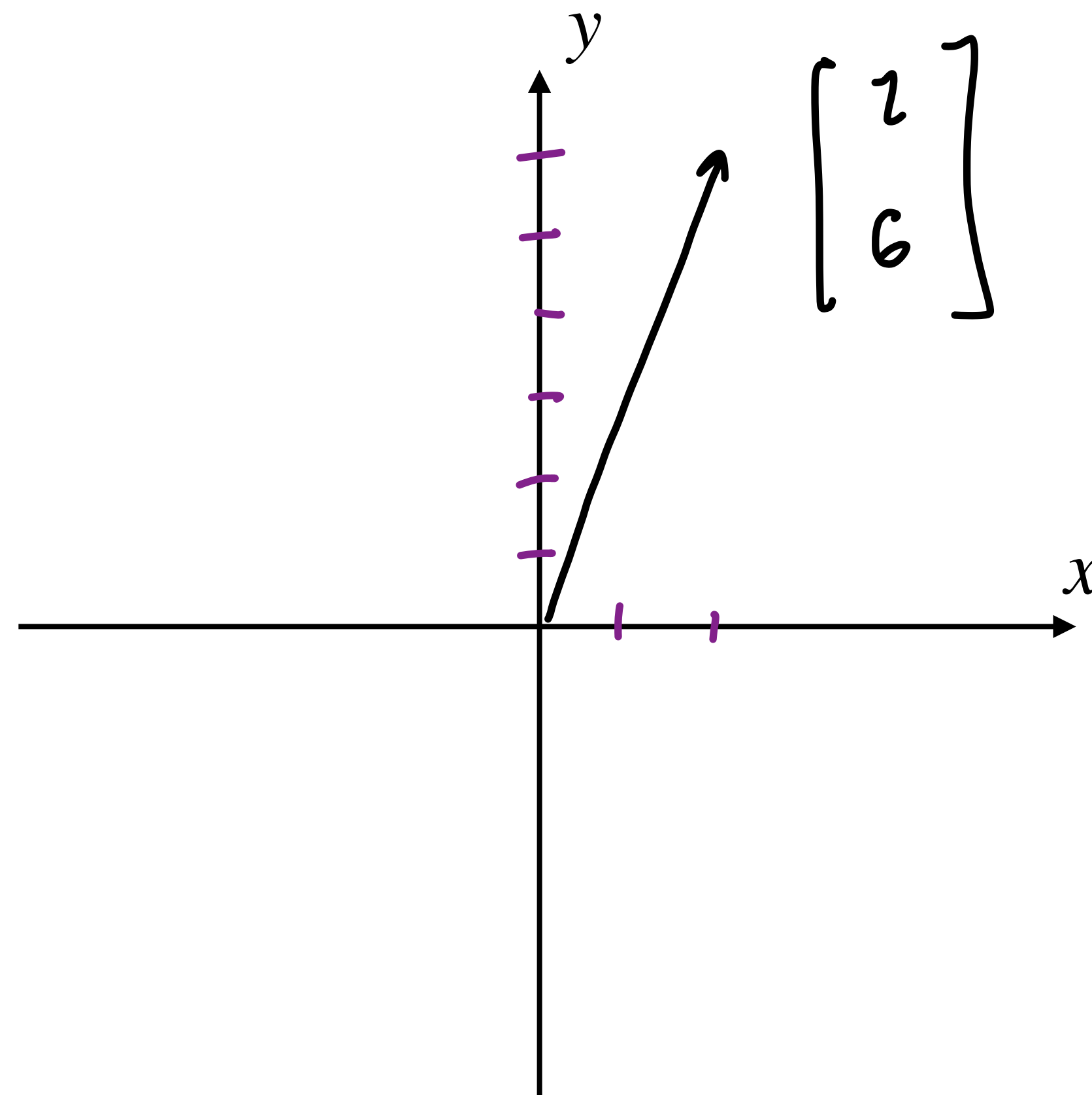
$$= \begin{bmatrix} x_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 2x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 \\ 2x_2 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \mathbf{x}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ 2x_2 \end{bmatrix}$$



Important

The vector may be a different
size after translation

Recall: Matrix-Vector Multiplication

matrix-vector multiplication only works if the number of *columns* of the matrix matches the dimension of the vector

$$\begin{array}{c} \textcolor{blue}{m} \left[\begin{array}{ccc} * & \dots & * \\ * & \dots & * \\ \vdots & \ddots & \vdots \\ * & \dots & * \\ * & \dots & * \end{array} \right] \quad \textcolor{red}{n} \left[\begin{array}{c} * \\ \vdots \\ * \end{array} \right] = \textcolor{blue}{m} \left[\begin{array}{c} * \\ * \\ \vdots \\ * \\ * \end{array} \right] \\ (m \times n) \quad \mathbb{R}^n \quad \mathbb{R}^m \end{array}$$

Motivating Questions

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What kind of functions can we define this way?

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New Interpretation

$$\vec{x} \mapsto A \vec{x}$$

$A\mathbf{x} = \mathbf{b}?$ $\equiv$ is there a vector which A transforms into $\mathbf{b}$?

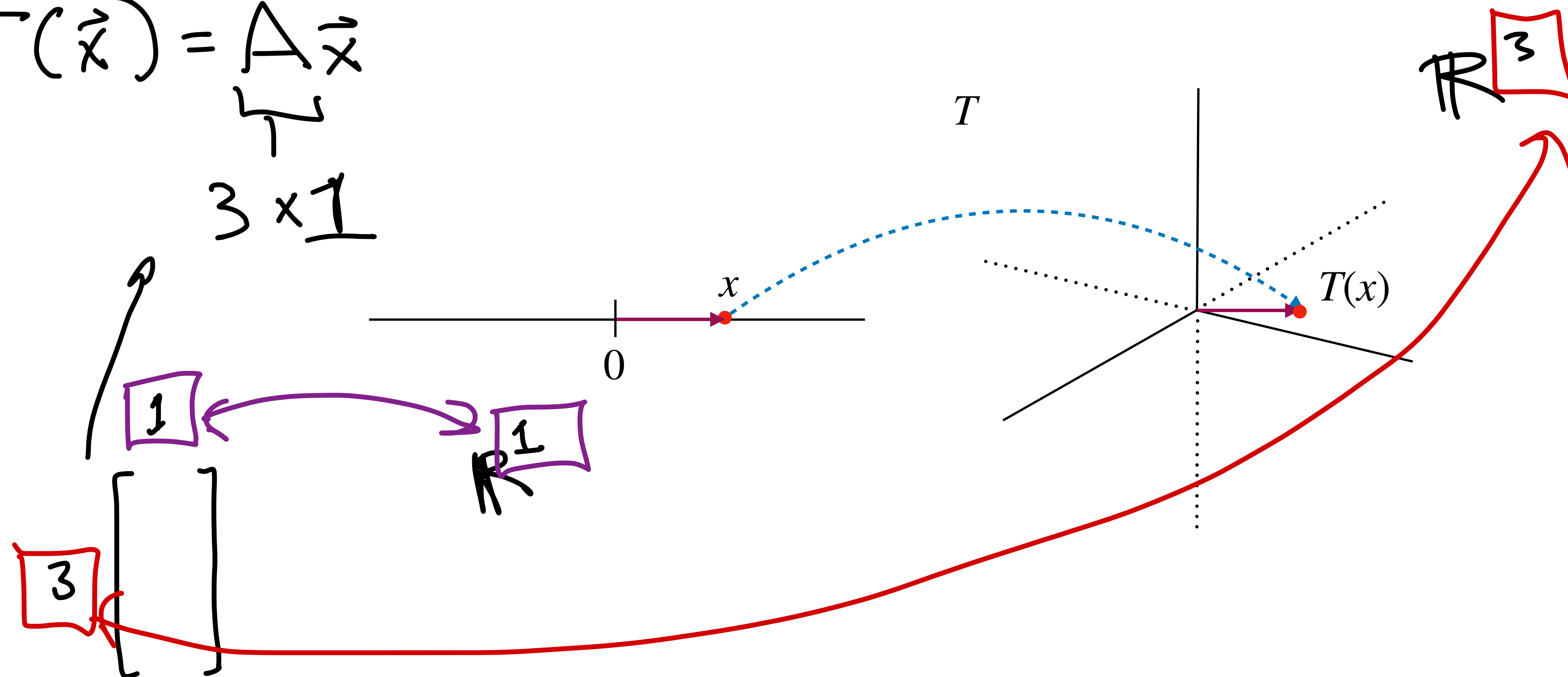
Solve $A\mathbf{x} = \mathbf{b}$ $\equiv$ find a vector which A transforms into $\mathbf{b}$

Question

Suppose a matrix transforms vectors according to the following picture. *What's the shape of the matrix?*

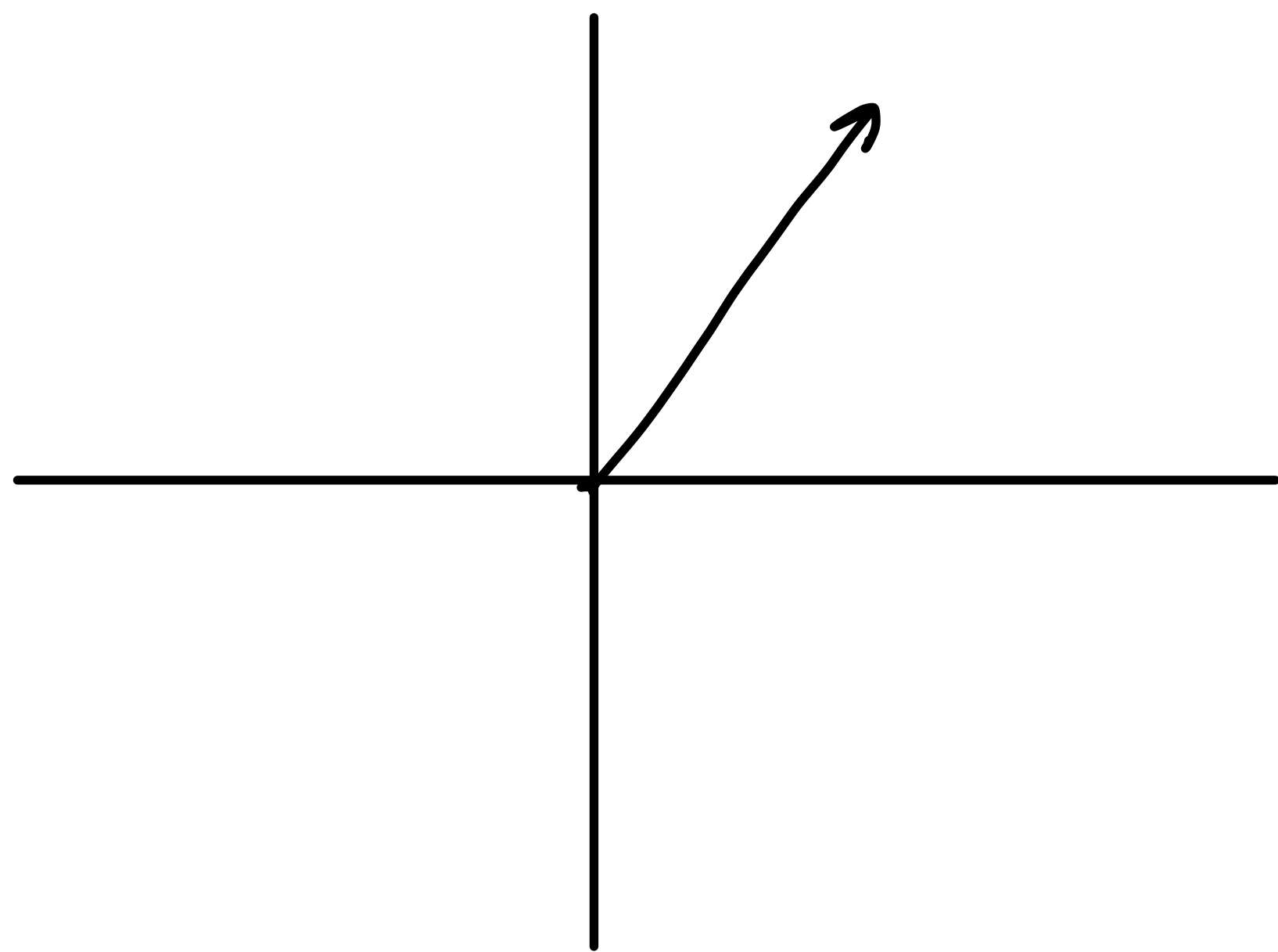
$$T(\vec{x}) = A \vec{x}$$

$\underbrace{\quad}_{3 \times 1}$

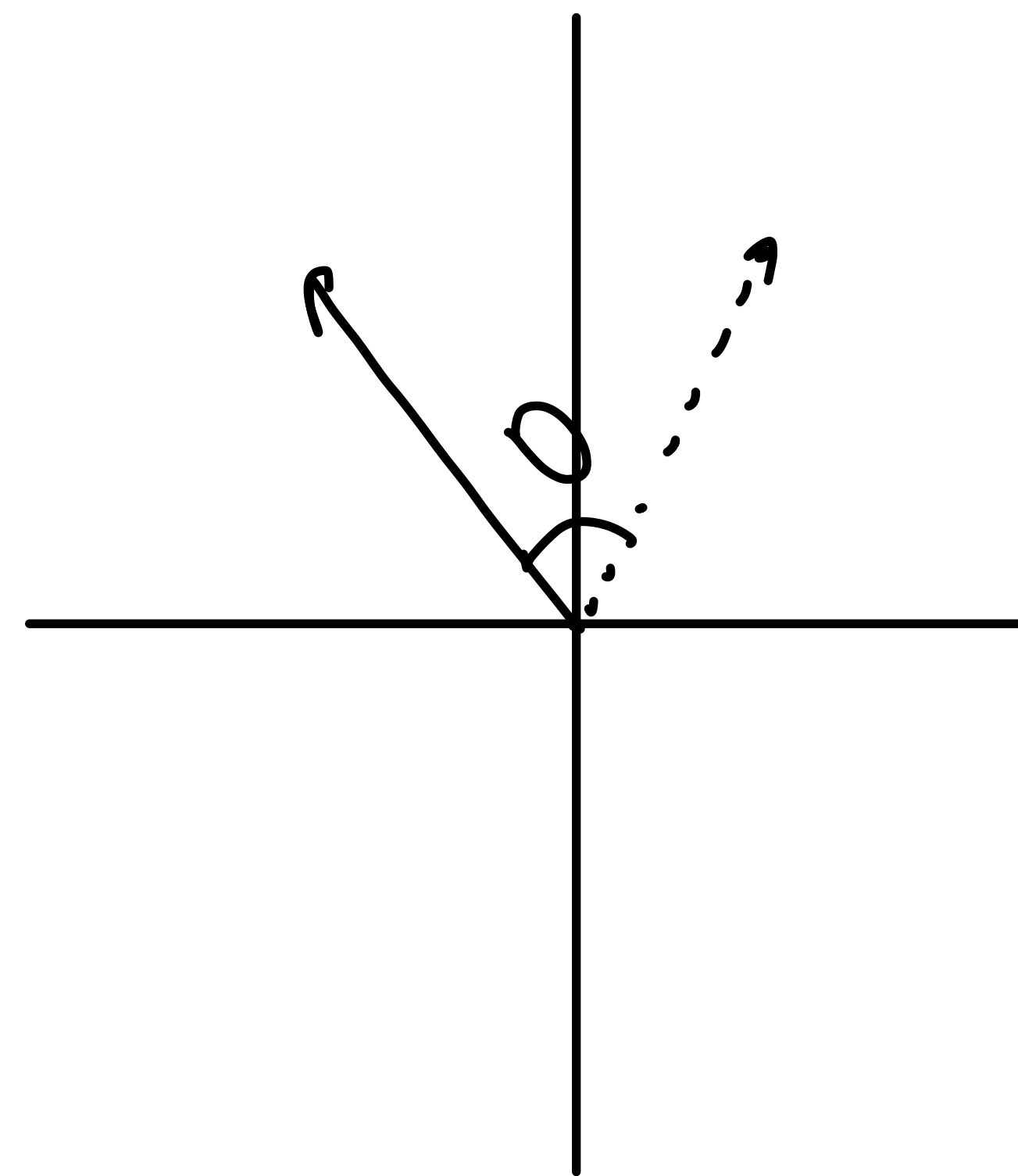


$$\mathbb{R}^n \rightarrow \mathbb{R}^n$$

Mapping between the same space can be viewed as a way of "moving around points"



rotation



Transformations

Transformations in General

Df. A *transformation* T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a function which maps every vector $\mathbf{v}$ in $\mathbb{R}^n$ to a vector $T(\mathbf{v})$ in $\mathbb{R}^m$

Transformations in General

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$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

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Df. A *transformation* T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a function which maps every vector $\mathbf{v}$ in $\mathbb{R}^n$ to a vector $T(\mathbf{v})$ in $\mathbb{R}^m$

$$T: \underbrace{\mathbb{R}^n}_{\substack{\text{domain} \\ \text{source space}}} \rightarrow \underbrace{\mathbb{R}^m}_{\substack{\text{codomain} \\ \text{target space}}}$$

Image and Range

Df. The image of $\vec{v}$ under T is $T(\vec{v})$

ex.

$$T(\vec{v}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \vec{v}$$

$\begin{bmatrix} 1 \\ 4 \end{bmatrix}$ is image of

Df. Set of all images under T is the range of T , $\text{ran}(T)$

ex.

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 \\ 2x_2 \end{bmatrix}$$

$$\text{ran}(T) = \mathbb{R}^2$$

$$\begin{bmatrix} 3 \\ 7 \end{bmatrix} = T\left(\begin{bmatrix} 3 \\ 7/2 \\ 0 \end{bmatrix}\right)$$

Image and Range

Df. For a vector $\mathbf{v}$, the *image* of $\mathbf{v}$ under the transformation T is the vector $T(\mathbf{v})$

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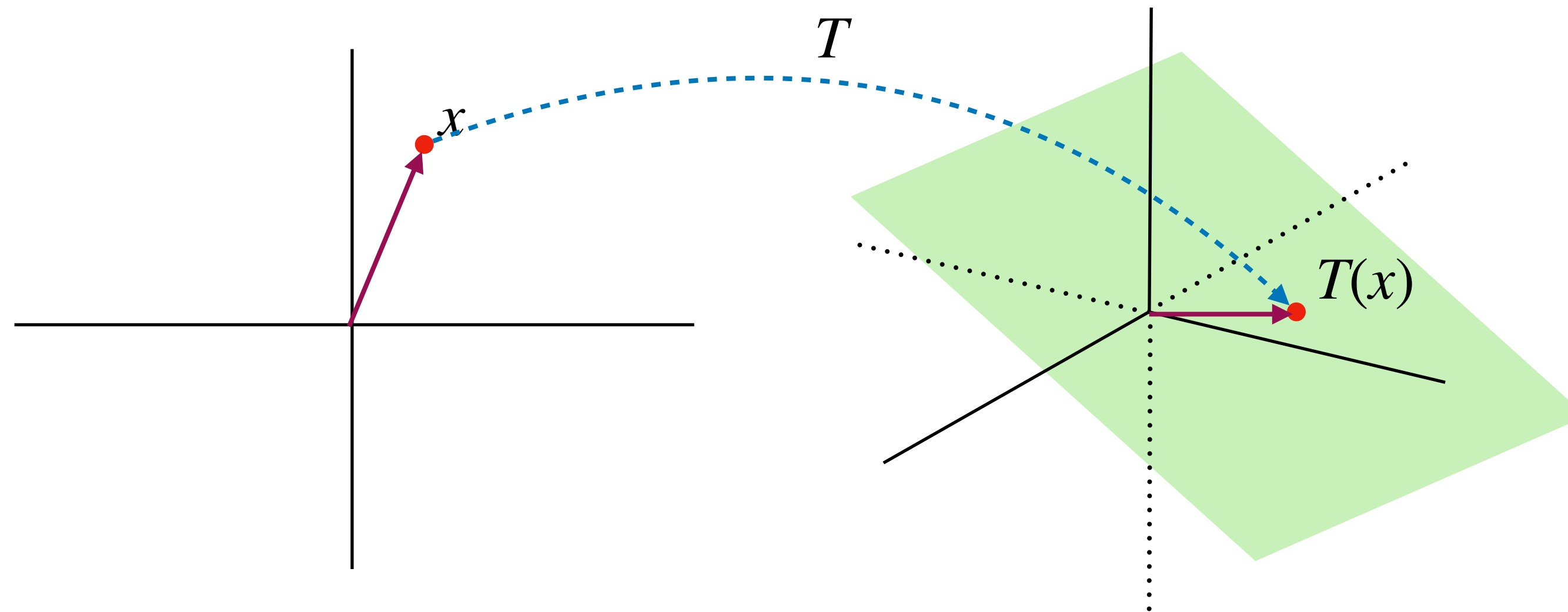
$$\text{ran}(T) = \{T(\mathbf{v}) : \mathbf{v} \in \mathbb{R}^n\}$$

image of $\mathbf{v}$ under $T \equiv$ output of T applied to $\mathbf{v}$

range of $T \equiv$ all possible output of T

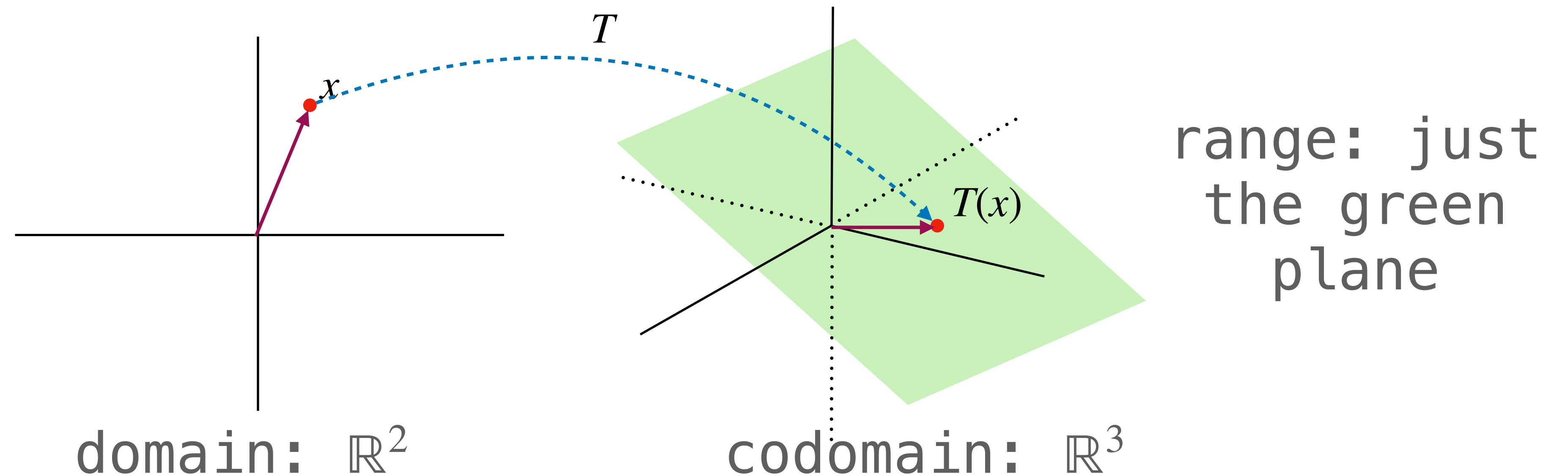
Codomain and Range

The codomain and range of a transformation may or may not be the same



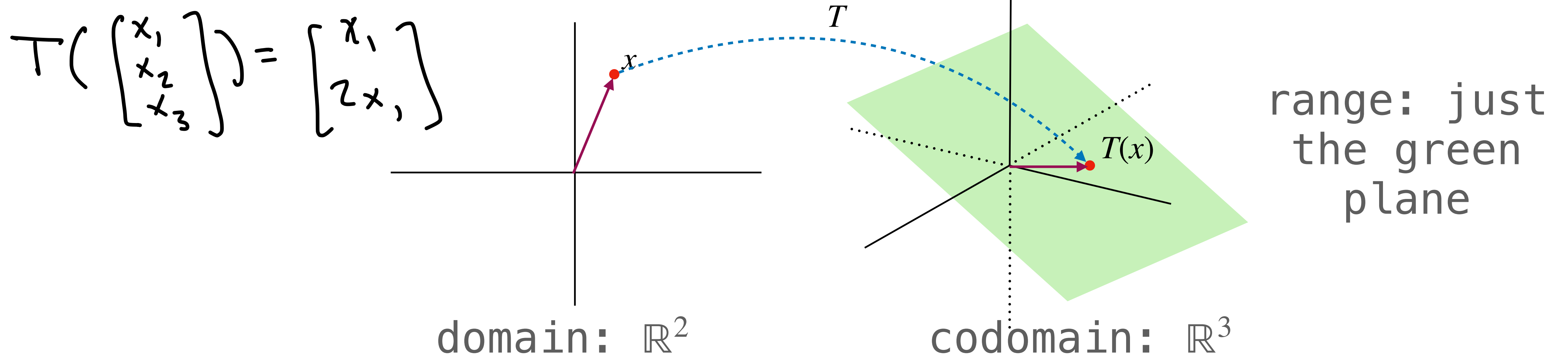
Codomain and Range

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Codomain and Range

The codomain and range of a transformation may or may not be the same



The range is always contained in the codomain

Example

T

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} x_1^2 \\ x_2 \\ 0 \end{bmatrix} \begin{bmatrix} \sqrt{5} \\ -3 \end{bmatrix} \mapsto \begin{bmatrix} 5 \\ -3 \\ 0 \end{bmatrix}$$

domain: $\mathbb{R}^2$

codomain: $\mathbb{R}^3$

$$\begin{aligned} \text{ran}(T) &= \left\{ T(\vec{v}) : \vec{v} \in \mathbb{R}^2 \right\} = \left\{ \begin{bmatrix} x_1^2 \\ x_2 \\ 0 \end{bmatrix} : x_1, x_2 \in \mathbb{R} \right\} \\ &= \left\{ v = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : v \in \mathbb{R}^3 \text{ s.t. } v_3 = 0, v_1 \geq 0 \right\} \end{aligned}$$

Matrix Transformations

Transformation of a Matrix

Transformation of a Matrix

Transformation of a Matrix

Df. The *transformation of a* $(m \times n)$ *matrix* A is the function $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$T(\mathbf{v}) = A\mathbf{v}$$

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i.e., Given $\mathbf{v}$, return A multiplied by $\mathbf{v}$

Transformation of a Matrix

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$$T(\mathbf{v}) = A\mathbf{v}$$

i.e., Given $\mathbf{v}$, return A multiplied by $\mathbf{v}$

e.g., $T(\mathbf{v}) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{v}$

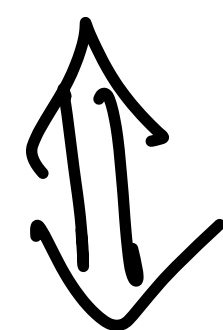
Range and Span

$$\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\} = \text{ran}([\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n])$$

||

$$\{[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] \vec{v} : \vec{v} \in \mathbb{R}^n\}$$

$$\text{ran}(\vec{x} \mapsto A\vec{x}) = \text{col}(\vec{x} \mapsto A\vec{x})$$



columns of A have

full

span $\Leftrightarrow$

1 pivot / row
in A

Linear Transformations

Motivating Questions

What kind of functions can we define this way?

How do we interpret what the transformation does to a set of vectors?

How does this relate to matrix equations?

Recall: Algebraic Properties of Matrix-Vector Multiplication

(additivity) $A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v}$

(homogeneity) $A(c\vec{v}) = c(A\vec{u})$

linearity is additivity + homogeneity

Recall: Algebraic Properties

Matrix-vector multiplication satisfies the following two properties:

1. $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$ (additivity)

2. $A(c\mathbf{v}) = c(A\mathbf{v})$ (homogeneity)

Example

Linear Transformations

Df. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is *linear* if it satisfies the following two properties

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (additivity)

2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (homogeneity)

Linear Transformations

Df. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is *linear* if it satisfies the following two properties

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (additivity)

2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (homogeneity)

Matrix transformations are linear transformations

Example: Identity

$$T(\mathbf{v}) = \mathbf{v}$$

add.

$$T(\vec{u} + \vec{v}) = \vec{u} + \vec{v} = T(\vec{u}) + T(\vec{v})$$

homogen.

$$T(c\vec{u}) = c\vec{u} = cT(\vec{u})$$

Linear

Example: Zero

Linear

$$T(\mathbf{v}) = \mathbf{0}$$

add.

$$T(\vec{v} + \vec{u}) = \vec{0}$$

=

$$\vec{0} = \vec{0} + \vec{0} = T(\vec{v}) + T(\vec{u}) \quad \checkmark$$

homogen.

$$T(c\vec{v}) = \vec{0}$$

=

$$\vec{0} = c\vec{0}$$

=

$$c T(\vec{v}) \quad \checkmark$$

$$T(\vec{v}) = \vec{1} \neq \vec{0}$$

add:

$$T(\vec{v} + \vec{u}) = \vec{1}$$

$\neq$

$$\vec{1} + \vec{1}$$

$$= T(\vec{v}) + T(\vec{u})$$

not linear

Example

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 + x_2 \\ x_2 - x_3 \\ 3x_1 \end{bmatrix}$$

add

$$T(\vec{v} + \vec{u}) =$$

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} + \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

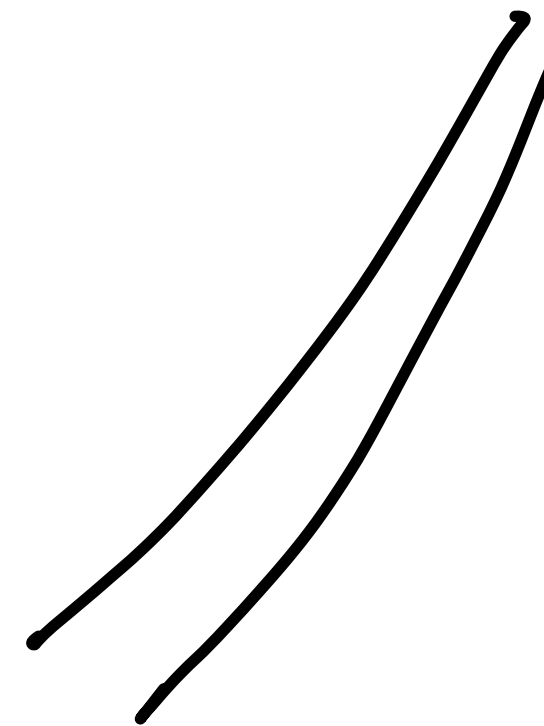
$$\begin{bmatrix} v_1 + u_1 \\ v_2 + u_2 \\ v_3 + u_3 \end{bmatrix}$$

$$\begin{bmatrix} (v_1 + u_1) + (v_2 + u_2) \\ (v_2 + u_2) - (v_3 + u_3) \\ 3(v_1 + u_1) \end{bmatrix}$$

"

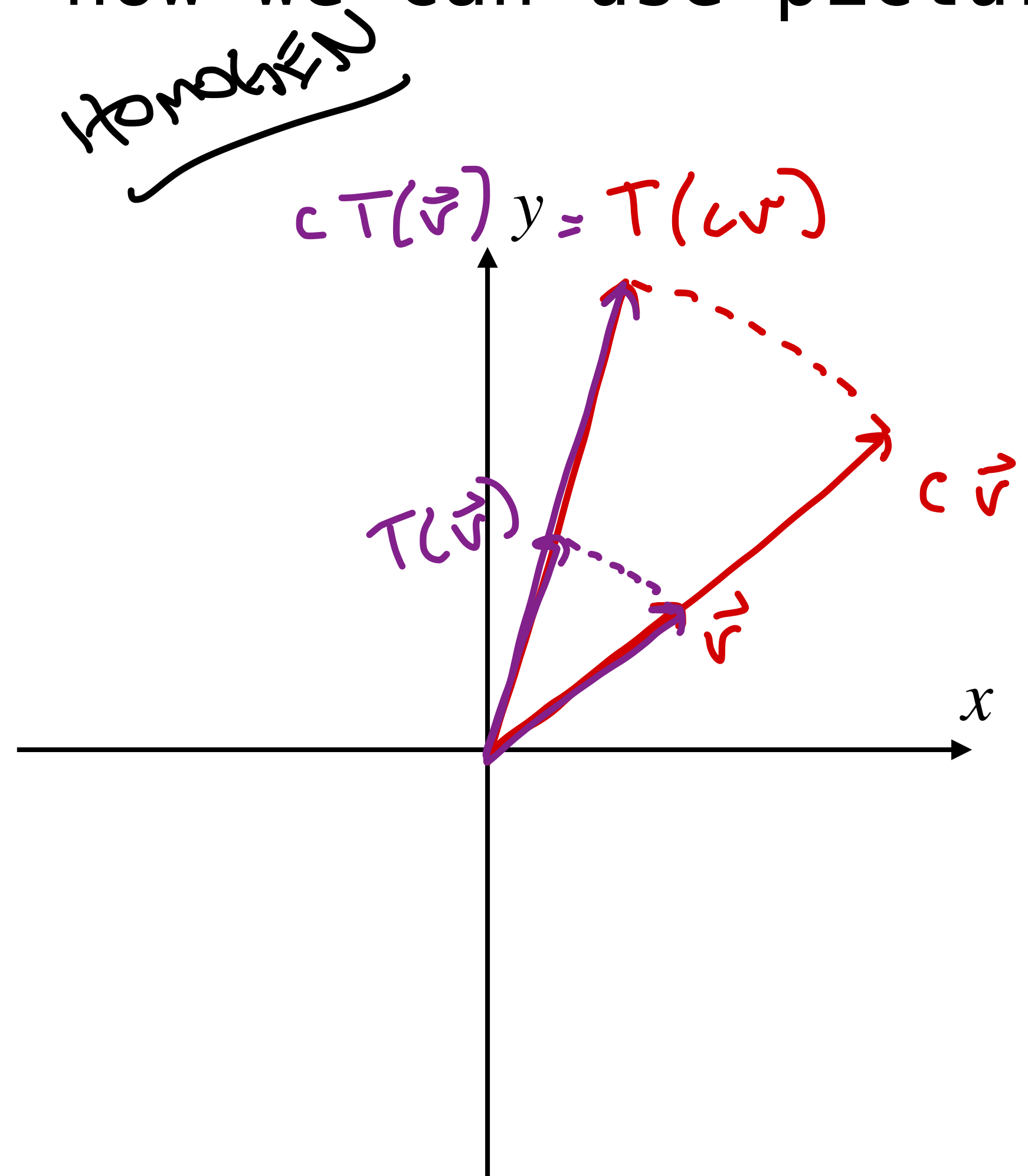
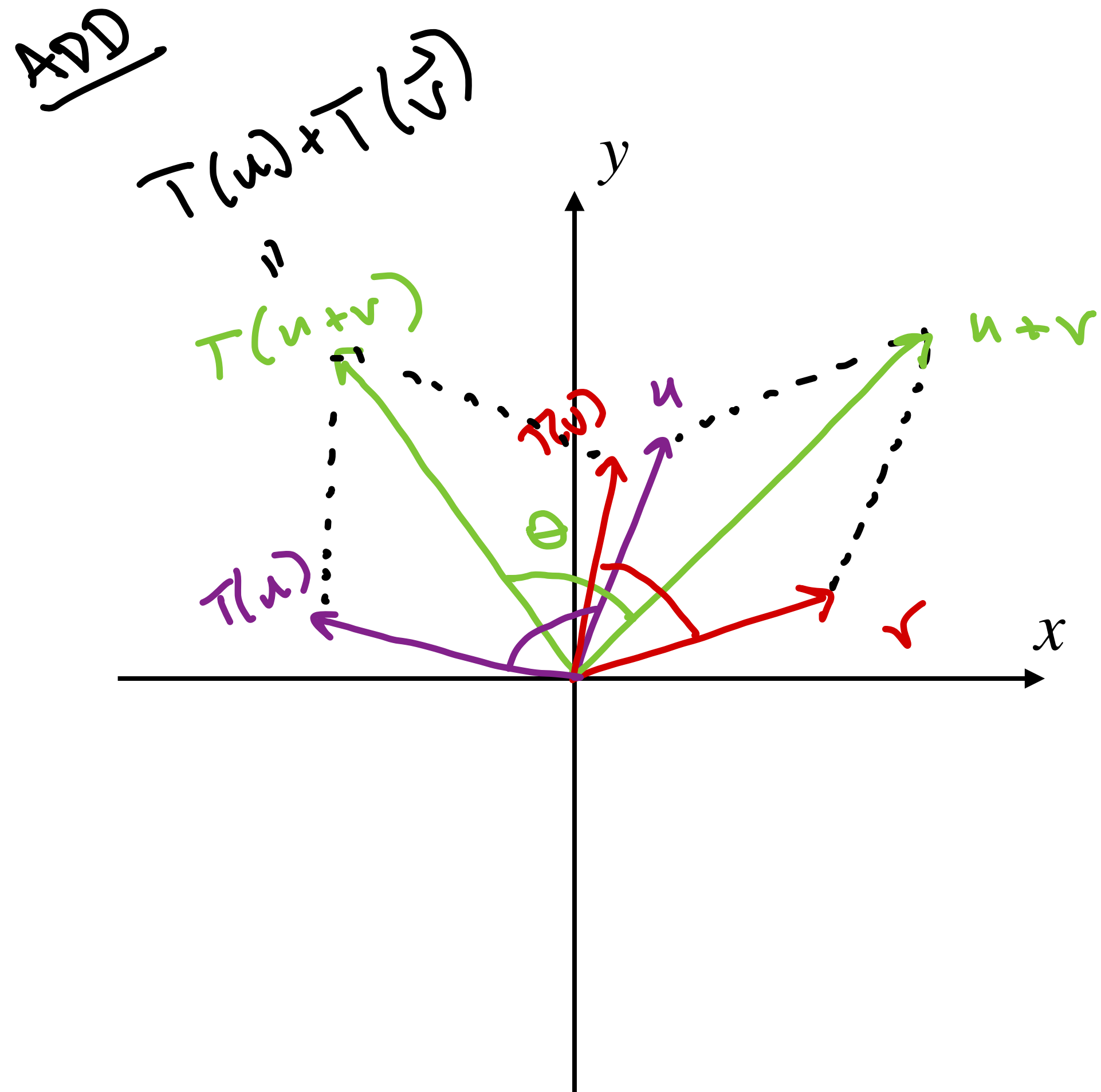
$$\begin{bmatrix} v_1 + v_2 \\ v_2 - v_3 \\ 3v_1 \end{bmatrix} + \begin{bmatrix} (u_1 + u_2) \\ (u_2 - u_3) \\ 3u_1 \end{bmatrix}$$

$$T(\vec{v}) + T(\vec{u})$$



Example: Rotation

We'll see this next time, but for now we can use pictures



Example: Indefinite Integrals

$$T(f) = \int f(x) dx$$

Disclaimer:
Advanced
Material

Example: Expectation

$$T(X) = \mathbb{E}[X]$$

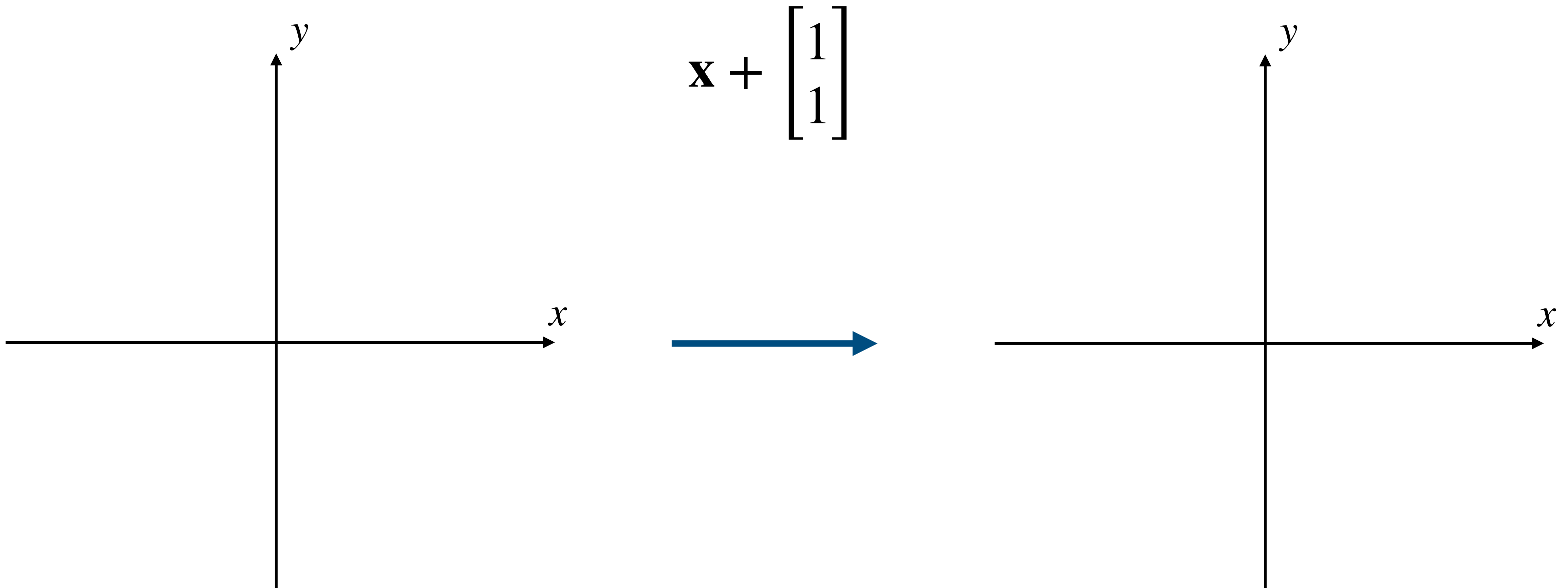
Disclaimer:
Advanced
Material

Non-Example: Squares

$$\begin{array}{ccc} T(x) & = & x^2 \\ \mathbb{R}^1 & & \mathbb{R}^1 \end{array}$$

$$\begin{aligned} T(6) &= T(2(3)) = 36 \\ 2T(3) &= 2(9) = \cancel{18}^4 \end{aligned}$$

Non-Example: Translation



Properties of Linear Transformations

The Zero Vector

The Zero Vector

$$T(\mathbf{0}) = ???$$

The Zero Vector

$$T(\mathbf{0}) = \mathbf{0}$$

The Zero Vector

$$T(\mathbf{0}) = \mathbf{0}$$

$$T(\vec{0}) = T(0\vec{v}) = 0T(\vec{v}) = \vec{0}$$

The zero vector is *fixed* by linear transformations

The Zero Vector

$$T(\mathbf{0}) = \mathbf{0}$$
Two red arrows originate from the text 'Note: These may be different dimensions!' below. One arrow points diagonally up and to the left towards the bold zero vector inside the parentheses of the equation $T(\mathbf{0}) = \mathbf{0}$. The other arrow points diagonally up and to the right towards the bold zero vector on the right side of the equation.

Note: These may be different dimensions!

The zero vector is *fixed* by linear transformations

A Single Condition

$$T(a\vec{r} + b\vec{u}) = aT(\vec{r}) + bT(\vec{u})$$

exercise: proof this cond. is equiv to
linearity

A Single Condition

$$T(a\mathbf{v} + b\mathbf{u}) = aT(\mathbf{v}) + bT(\mathbf{u})$$

A Single Condition

$$T(a\mathbf{v} + b\mathbf{u}) = aT(\mathbf{v}) + bT(\mathbf{u})$$

We can combine our linearity conditions

A Single Condition

Thm. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is linear if and only if for any vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^m$ and any real numbers a and b ,

$$T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$$

A Single Condition

Thm. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is linear if and only if for any vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^m$ and any real numbers a and b ,

$$T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$$

It's often easiest to show this
single condition

Linear Combinations

$$T\left(\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n\right) =$$

$$T\left(\alpha_1 \vec{v}_1\right) + \dots + T\left(\alpha_n \vec{v}_n\right) =$$

$$\alpha_1 T\left(\vec{v}_1\right) + \dots + \alpha_n T\left(\vec{v}_n\right) =$$

Geometry of Matrix Transformations

Motivating Questions

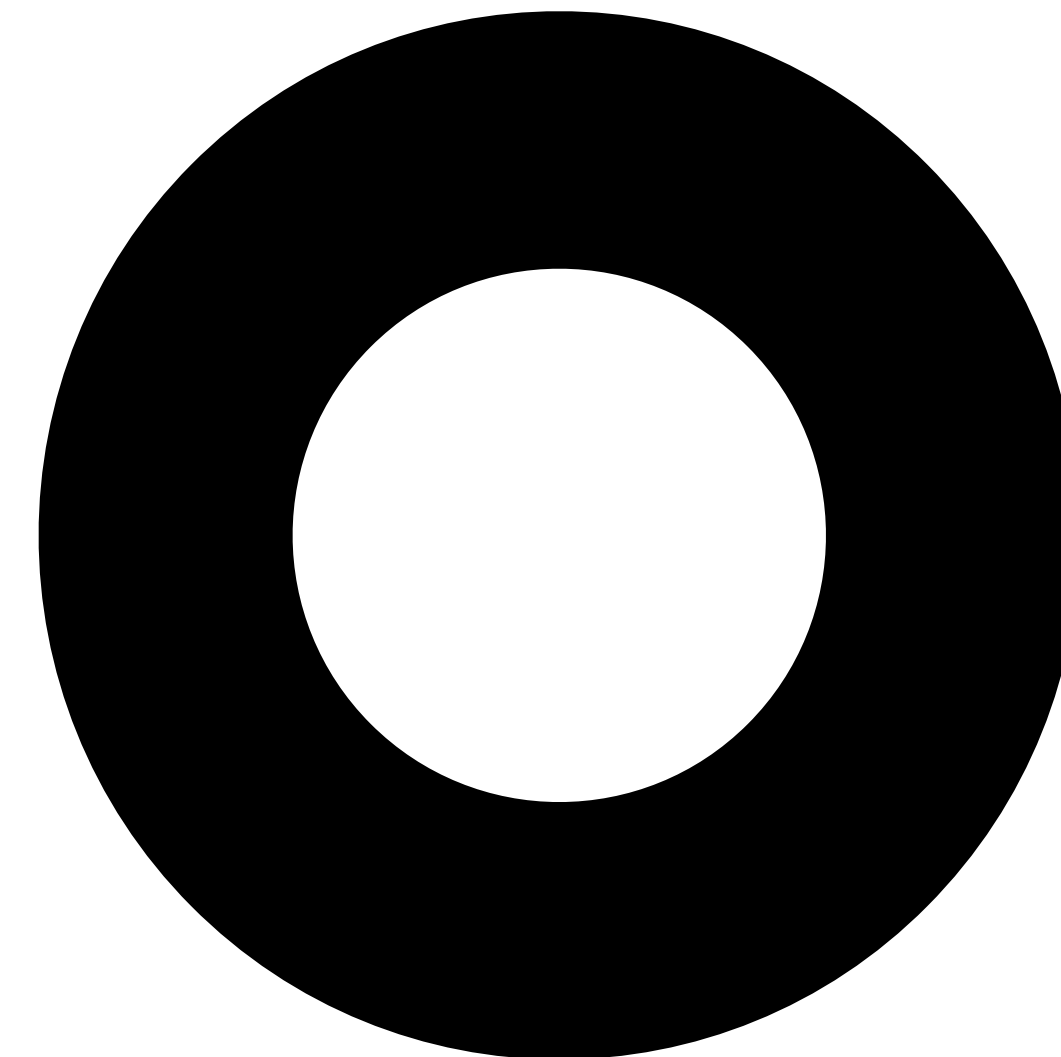
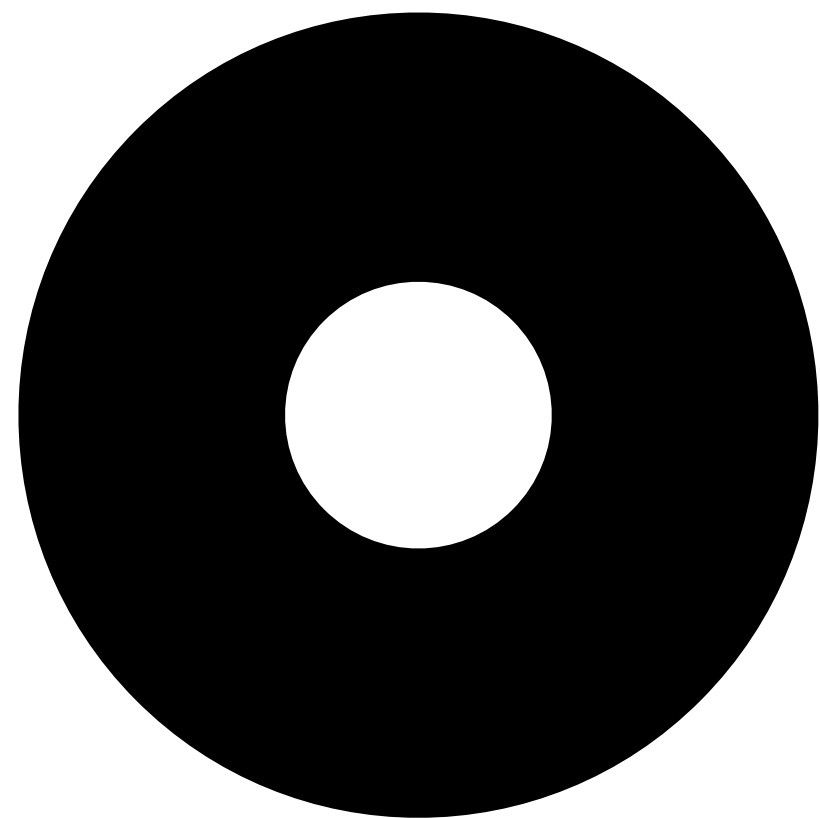
What kind of functions can we define this way?

How do we interpret what the transformation does to a set of vectors?

How does this relate to matrix equations?

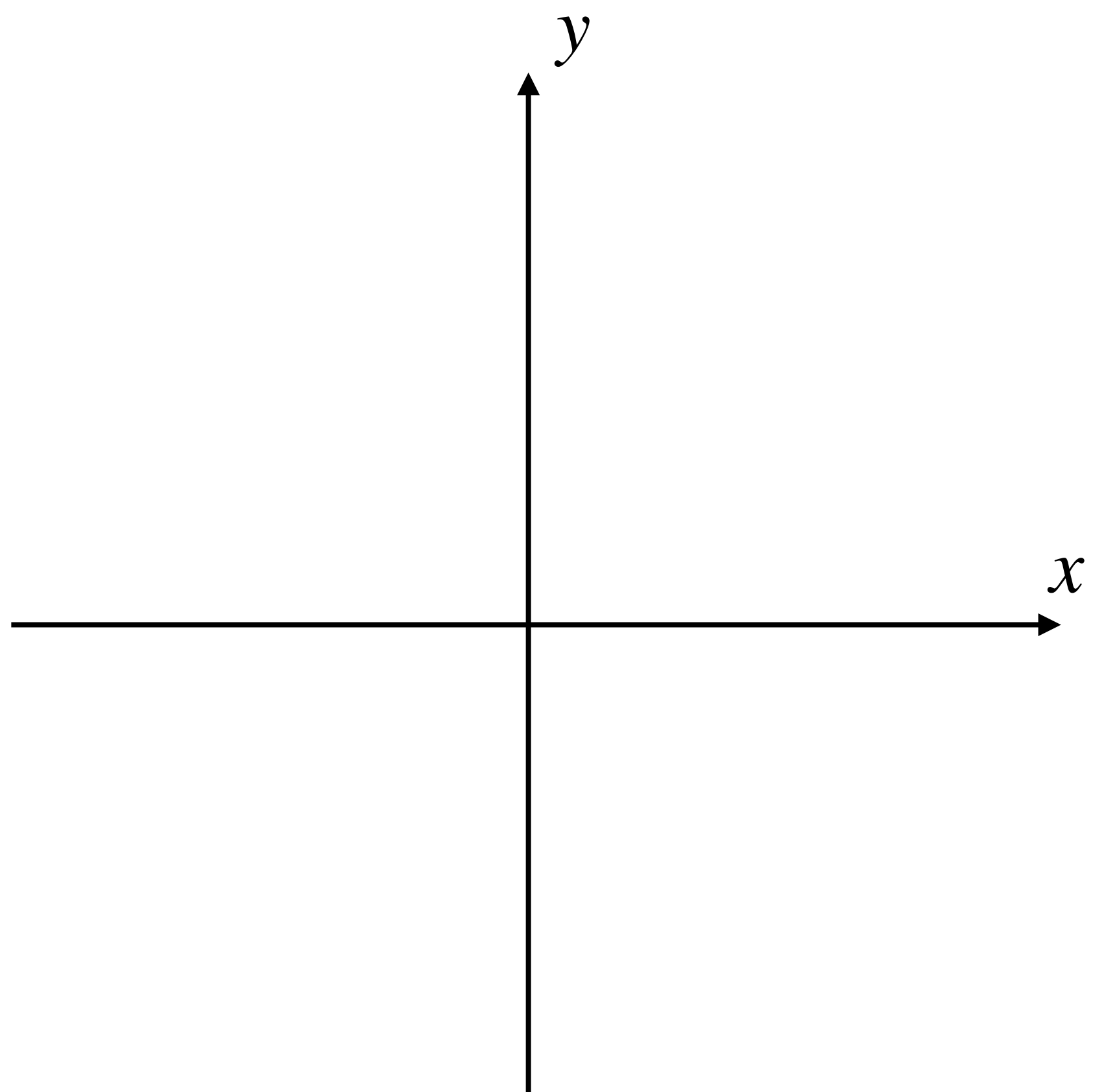
Matrix transformations change the
"shape" of a set of set of
vectors (points)

Example: Dilation

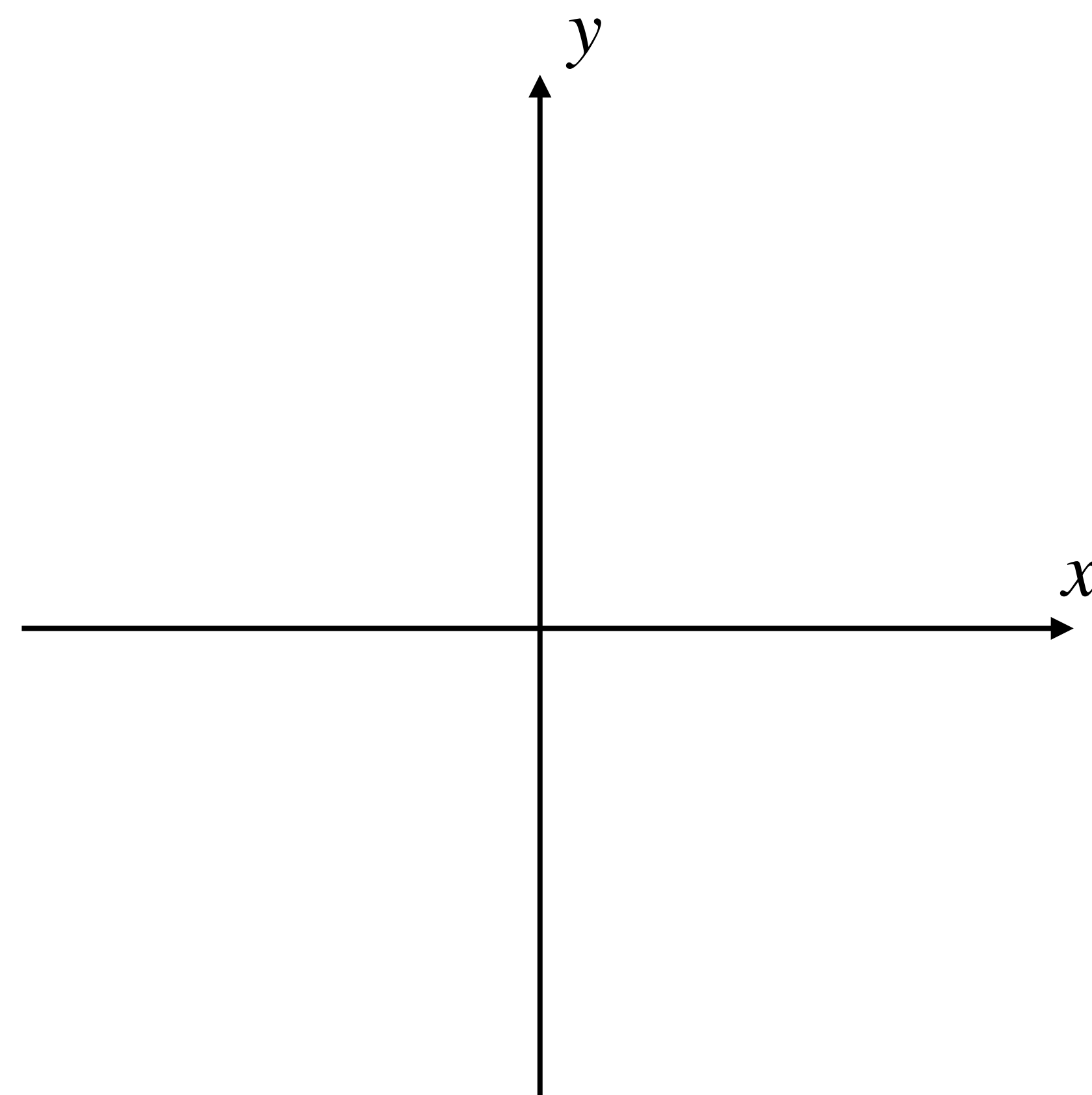


Example: Dilation

$$\begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} rx_1 \\ rx_2 \end{bmatrix}$$

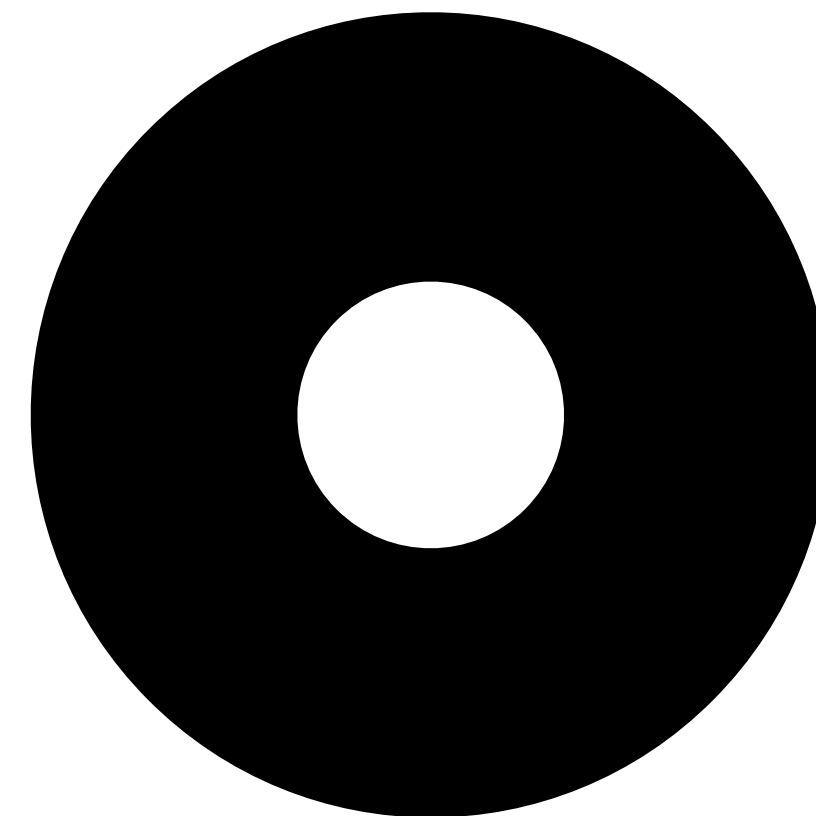
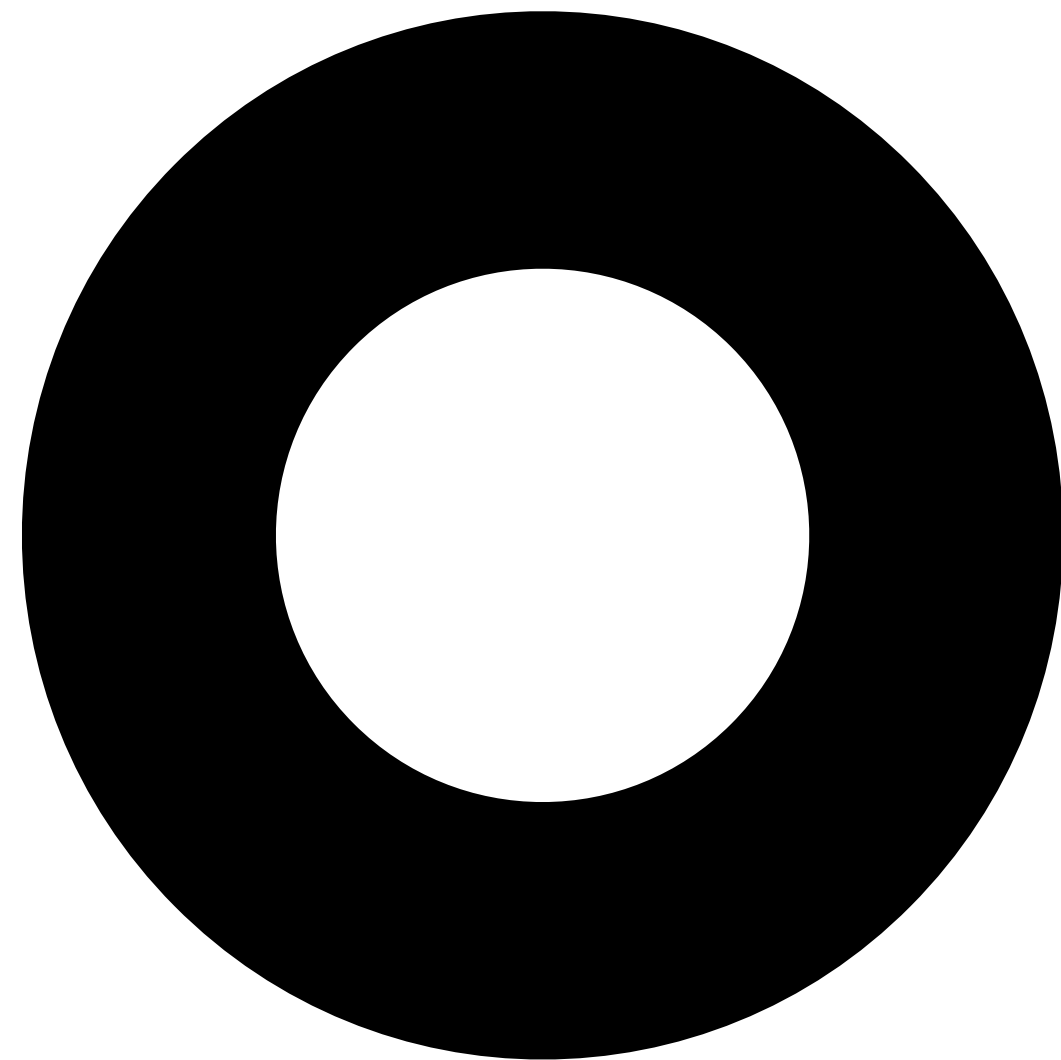


$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{x}$$

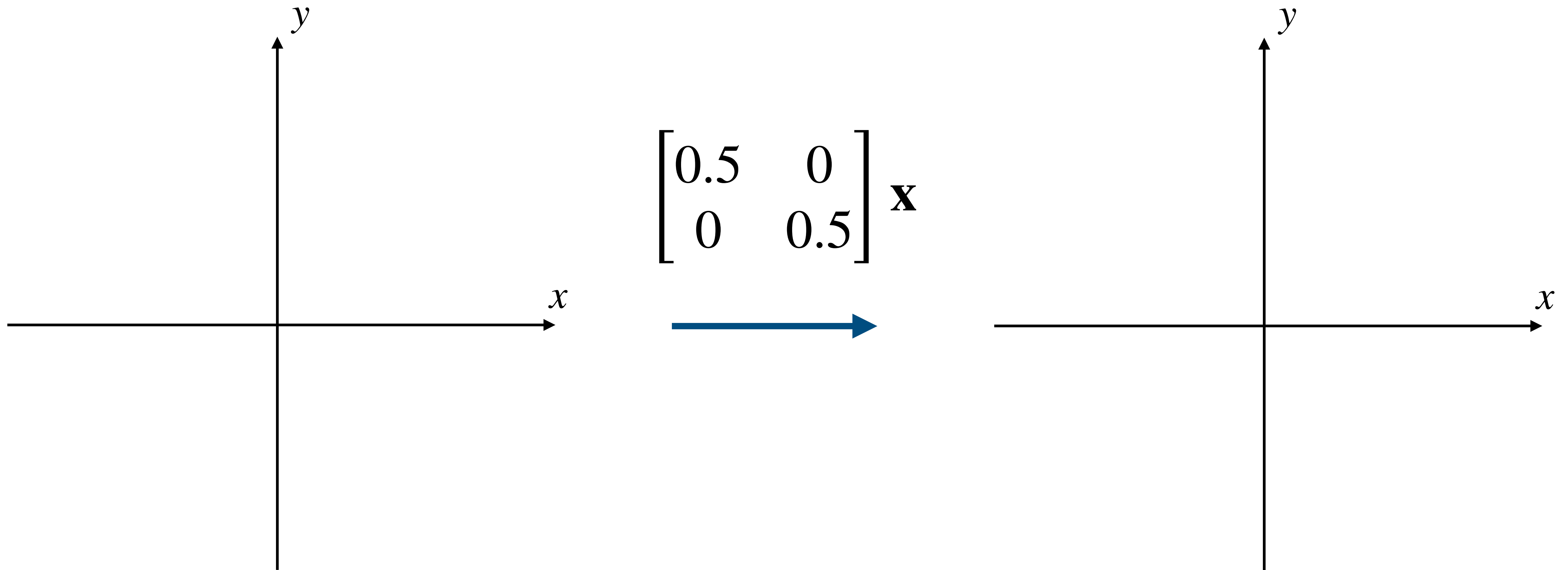
if $r > 1$, then the transformation pushes points away from the origin

Example: Contraction



Example: Contraction

$$\begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} rx_1 \\ rx_2 \end{bmatrix}$$



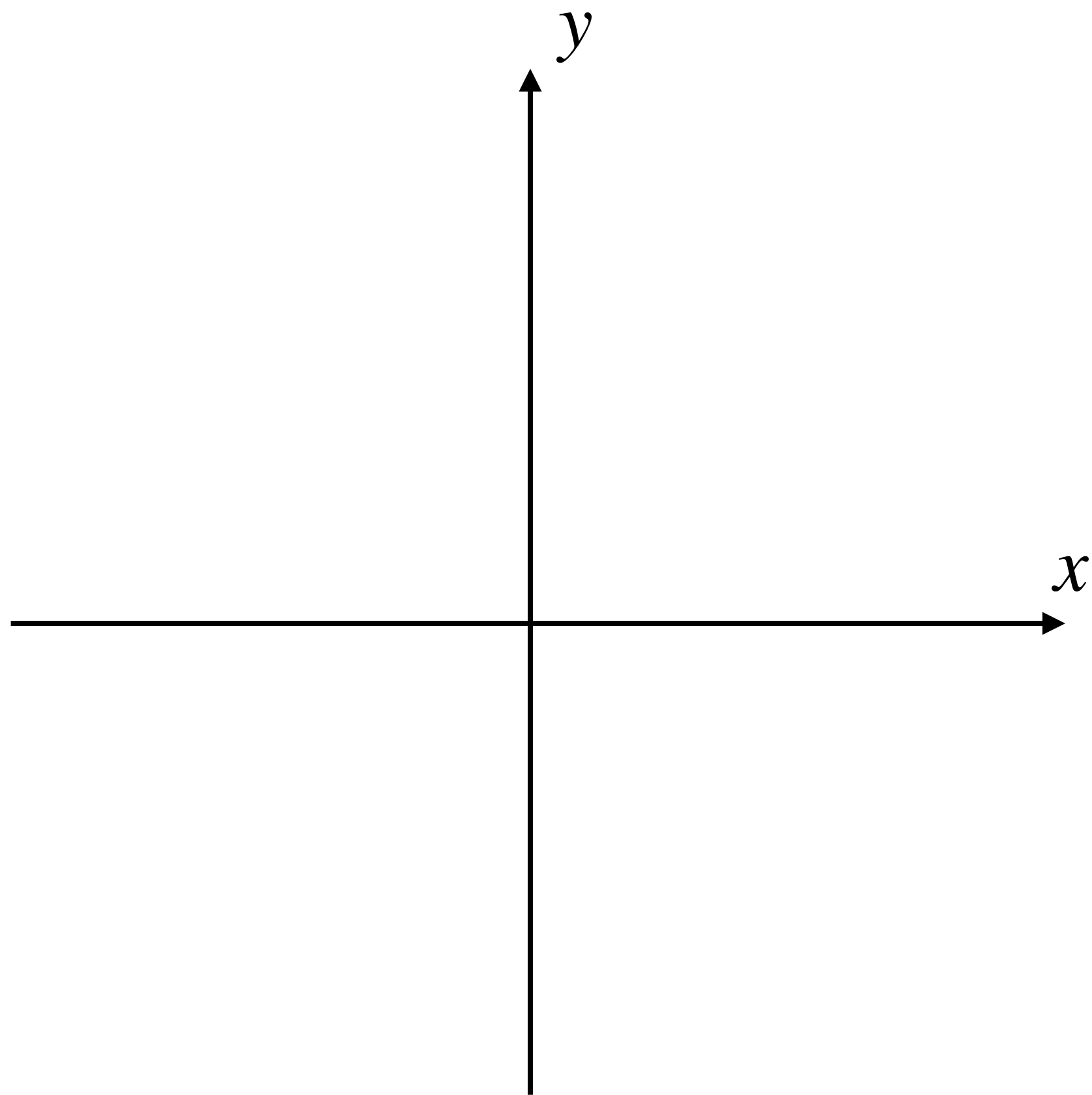
if $0 \leq r \leq 1$, then the transformation pulls points towards the origin

Example: Shearing

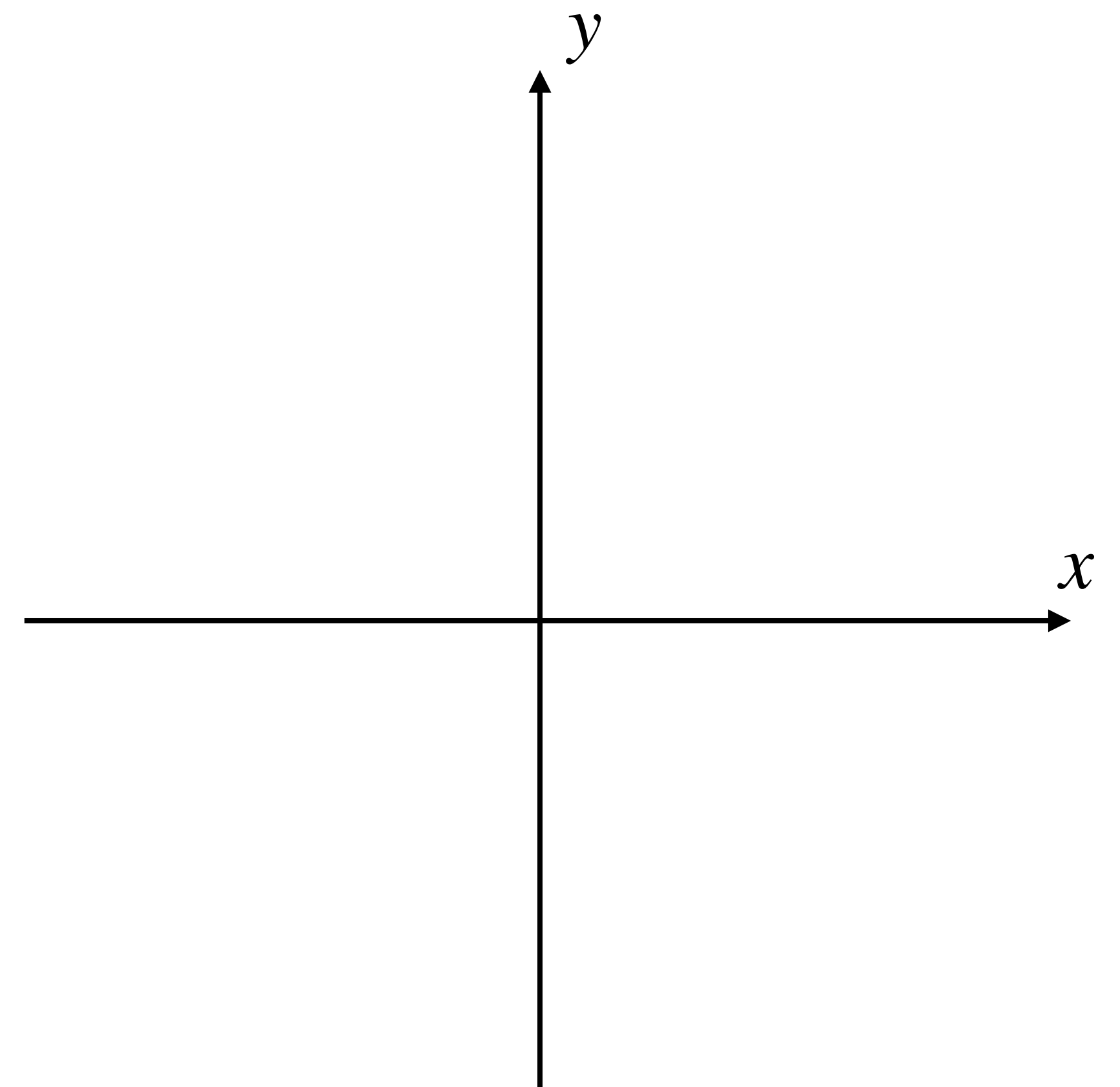


Example: Shearing

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_2 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x}$$

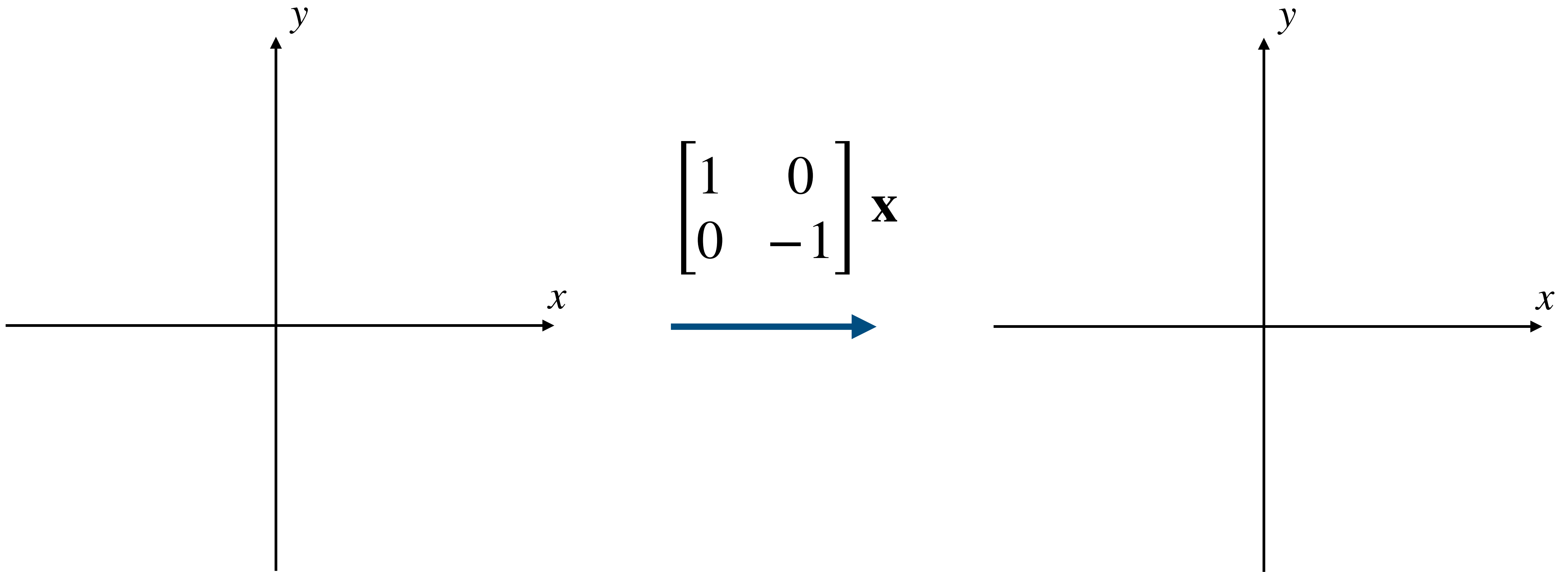



Question

$$\mathbf{x} \mapsto \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x}$$

Draw how this matrix transforms points. What kind of transformation does it represent?

Answer: Reflection



demo

Summary

Matrices can be viewed as **linear transformations**

Matrix transformations change the **shape** of points sets

Linear transformations behave well with respect to **linear combinations**